ACM/IDS 104 APPLIED LINEAR ALGEBRA
PRACTICE PROBLEMS FOR LECTURE 14: SOLUTIONS

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Problem 14A. Principal Component Analysis: Toy Example
As a friendly reminder, whenever we learn (or develop) a new method or algorithm, it is always very useful to explore how the method works in simple, particular cases where the solution to the problem the algorithm is intended to solve is known. This illuminates the algorithm, makes it more intuitively clear, and serves as a common-sense check.

Consider the data consisting of six points in the plane shown in the left panel of Fig. 1. This data set is inspired by the constellation Lyra shown in the right panel of the figure. Compose the data matrix, find the principal components of the data (by hand), and sketch them on the scatterplot.

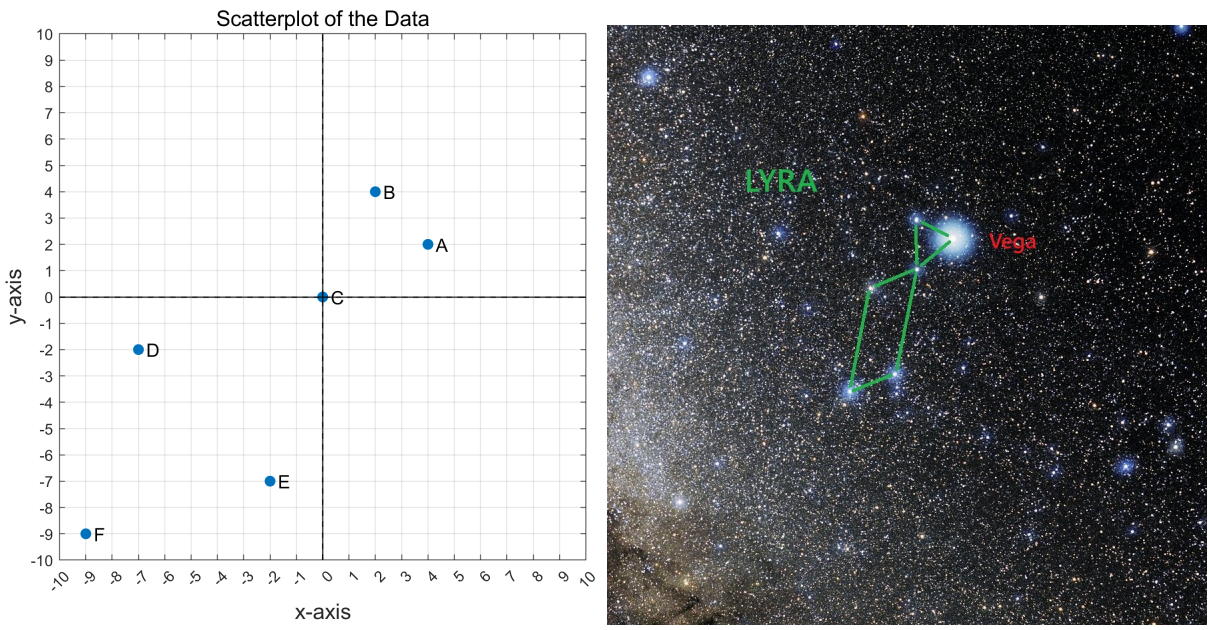

Figure 1. Left panel: Scatterplot of the data. Right panel: Constellation Lyra (Photo by Eckhard Slawik, `https://noirlab.edu/public/education/constellations/lyra/`; minor edits by KZ.). Lyra is a small constellation located in the Northern Hemisphere. The brightest star in Lyra is Vega, which is the fifth-brightest star in the sky. Vega was the pole star in the year 12,000 BC, and, because of the precession of Earth's rotational axis, will again become the pole star around 14,000 AD. In Greek mythology, Lyra represents the lyre of Orpheus. In Wales, Lyra is known as King David's harp.

Solution:
The data consists of $n = 6$ observations A, B, C, D, E, F in $\mathbb{R}^{m=2}$. The data matrix is the 2×6 matrix, whose columns are the coordinates of the observations:

$$X = \begin{bmatrix} 4 & 2 & 0 & -2 & -7 & -9 \\ 2 & 4 & 0 & -7 & -2 & -9 \end{bmatrix}. \tag{1}$$

The row-mean vector is

$$\mu_X = \begin{bmatrix} -2 \\ -2 \end{bmatrix}. \tag{2}$$

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We can compute the covariance matrix C_X directly via $C_X = \frac{1}{n}XX^T - \mu_X\mu_X^T$, Alternatively, we can first center the data at zero by replacing each column x_j of X with $x_j - \mu_X$,

$$X \to \tilde{X} = X - [\mu_X, \ldots, \mu_X] = \begin{bmatrix} 6 & 4 & 2 & 0 & -5 & -7 \\ 4 & 6 & 2 & -5 & 0 & -7 \end{bmatrix}, \tag{3}$$

so that $\mu_{\tilde{X}} = 0$, and then computing the covariance matrix via $C_X = \frac{1}{n}\tilde{X}\tilde{X}^T$. Either way, the covariance matrix is

$$C_X = \frac{1}{6}\begin{bmatrix} 130 & 101 \\ 101 & 130 \end{bmatrix}. \tag{4}$$

The principal components of X are the orthonormal eigenvectors of C_X. The eigenvalues of C_X are the roots of the characteristic polynomial

$$p(\lambda) = \det(C_X - \lambda I) = \left(\frac{1}{6}\right)^2 \det\begin{bmatrix} 65-6\lambda & 101 \\ 101 & 65-6\lambda \end{bmatrix} = \left(\frac{1}{6}\right)^2 \left((65-6\lambda)^2 - 101^2\right), \tag{5}$$

which are

$$\lambda = \frac{130 \pm 101}{6} \quad \Rightarrow \quad \lambda_1 = \frac{231}{6} \quad \text{and} \quad \lambda_2 = \frac{29}{6}. \tag{6}$$

The corresponding eigenvectors are:

$$\begin{aligned} \lambda_1 = \frac{231}{6}: \quad & (C_X - \lambda_1 I)v_1 = 0 \;\Leftrightarrow\; \frac{1}{6}\begin{bmatrix} -101 & 101 \\ 101 & -101 \end{bmatrix} v_1 = 0 \;\Leftrightarrow\; v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \\ \lambda_2 = \frac{29}{6}: \quad & (C_X - \lambda_2 I)v_2 = 0 \;\Leftrightarrow\; \frac{1}{6}\begin{bmatrix} 101 & 101 \\ 101 & 101 \end{bmatrix} v_2 = 0 \;\Leftrightarrow\; v_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix} \end{aligned} \tag{7}$$

The principal components of the data are, thus,

$$q_1 = \frac{v_1}{\|v_1\|} = \begin{bmatrix} \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{bmatrix} \quad \text{and} \quad q_2 = \frac{v_2}{\|v_2\|} = \begin{bmatrix} -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{bmatrix}. \tag{8}$$

Figure 2 shows the data together with its principal components. The dispersion (variability) of the data is the largest along the first principal component and the smallest along the second one.

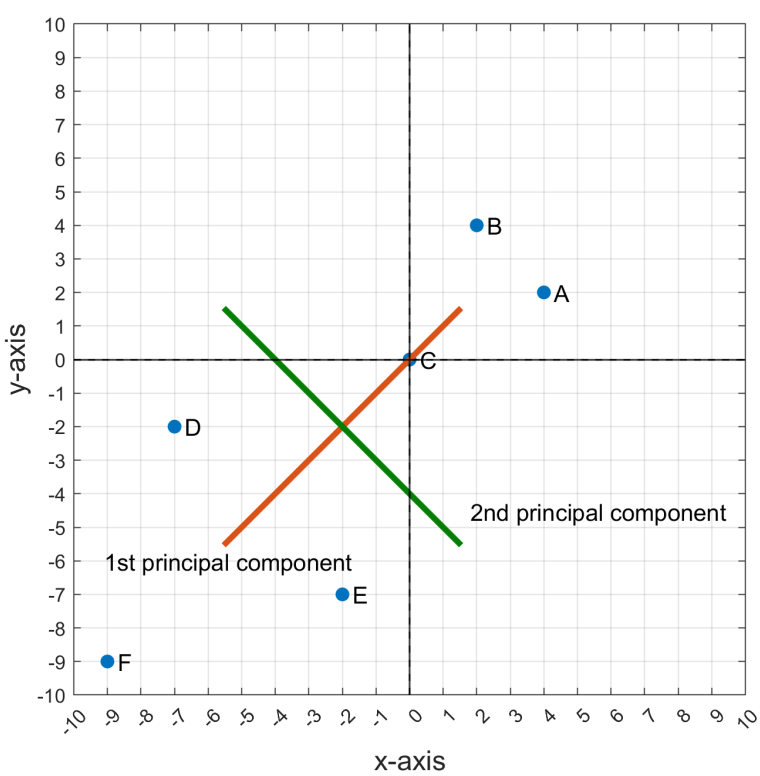

FIGURE 2. The data and its principal components

Problem 14B. Total Variance of the Data Matrix

Let X be an $m \times n$ data matrix, where columns $x_1, \ldots, x_n \in \mathbb{R}^m$ represent measurements (or observations) and rows $x^1, \ldots, x^m \in \mathbb{R}^n$ represent features (or variables). Suppose that X is centered, that is, each row has zero mean. Geometrically, this means that the cloud of points $x_1, \ldots, x_n \in \mathbb{R}^m$ is centered at the origin. Then, the variance of the ith feature is[1]

$$\mathbb{V}(x^i) = \frac{1}{n}\sum_{j=1}^{n} x_{ij}^2, \quad i = 1, \ldots, m. \tag{9}$$

The *total variance* $\mathbb{V}(X)$ of the data matrix X is defined as the sum of variances of all features.

(a) Let A be an $m \times n$ matrix and B be an $n \times m$ matrix. Show that

$$\operatorname{tr}(AB) = \operatorname{tr}(BA). \tag{10}$$

This means that, although matrices generally do not commute, $AB \neq BA$, they always commute under the trace. This property makes the trace very interesting algebraically, and it is used a lot in a wide range of computations, proofs, and applications.

(b) Prove that the total variance equals the trace of the covariance matrix C_X of the data matrix,

$$\mathbb{V}(X) = \operatorname{tr}(C_X), \qquad \text{where } C_X = \frac{1}{n}XX^T. \tag{11}$$

(c) Let $\lambda_1, \ldots, \lambda_m$ be the eigenvalues of the covariance matrix C_X. Prove that

$$\mathbb{V}(X) = \sum_{i=1}^{m} \lambda_i. \tag{12}$$

Solution:

(a) We can prove this property using the definition of trace and matrix multiplication:

$$\begin{aligned} \operatorname{tr}(AB) &= \sum_{i=1}^{m}(AB)_{ii} = \sum_{i=1}^{m}\left(\sum_{j=1}^{n} a_{ij}b_{ji}\right) \\ \operatorname{tr}(BA) &= \sum_{j=1}^{n}(BA)_{jj} = \sum_{j=1}^{n}\left(\sum_{i=1}^{m} b_{ji}a_{ij}\right) = \sum_{i=1}^{m}\left(\sum_{j=1}^{n} a_{ij}b_{ji}\right) \end{aligned} \quad \Rightarrow \quad \operatorname{tr}(AB) = \operatorname{tr}(BA). \tag{13}$$

(b) The total variance of X is

$$\mathbb{V}(X) = \sum_{i=1}^{m} V(x^i) = \frac{1}{n}\sum_{i=1}^{m}\sum_{j=1}^{n} x_{ij}^2. \tag{14}$$

Recall that in Practice Problem 4B, we defined the Frobenius norm of a matrix A as

$$\|A\|_F = \sqrt{\operatorname{tr}(A^TA)}, \tag{15}$$

and we have shown that

$$\|A\|_F = \sqrt{\sum_{i=1}^{m}\sum_{j=1}^{n} a_{ij}^2}. \tag{16}$$

This means that the total variance is simply the rescaled squared Frobenius norm:

$$\mathbb{V}(X) = \frac{1}{n}\|X\|_F^2. \tag{17}$$

On the other hand, using $\operatorname{tr}(AB) = \operatorname{tr}(BA)$ proved in part (a), the trace of the covariance matrix is

$$\operatorname{tr}(C_X) = \operatorname{tr}\left(\frac{1}{n}XX^T\right) = \frac{1}{n}\operatorname{tr}\left(XX^T\right) = \frac{1}{n}\operatorname{tr}\left(X^TX\right) = \frac{1}{n}\|X\|_F^2. \tag{18}$$

Therefore,

$$\mathbb{V}(X) = \operatorname{tr}(C_X) = \frac{1}{n}\|X\|_F^2. \tag{19}$$

[1] If X is not centered, then the variance is $\mathbb{V}(x^i) = \frac{1}{n}\sum_{j=1}^{n} x_{ij}^2 - \left(\frac{1}{n}\sum_{j=1}^{n} x_{ij}\right)^2$, which is less convenient to work with.

(c) Since C_X is symmetric, by the Spectral Theorem, it can be written as $C_X = Q\Lambda Q^T$, where Λ is the diagonal matrix of eigenvalues and Q is the orthogonal matrix of eigenvectors (principal components of X). Using the property $\mathrm{tr}(AB) = \mathrm{tr}(BA)$ proved in part (a), we have:

$$\mathbb{V}(X) = \mathrm{tr}(C_X) = \mathrm{tr}\left(Q\Lambda Q^T\right) = \mathrm{tr}\left(Q^T Q\Lambda\right) = \mathrm{tr}\left(I\Lambda\right) = \mathrm{tr}\left(\Lambda\right) = \sum_{i=1}^{m} \lambda_i. \tag{20}$$

In fact, for any (not only symmetric) square matrix A, its trace equals the sum of its eigenvalues. It is just that the proof in this case is a bit longer than the above proof for symmetric matrices.